



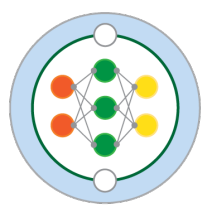
Learning Propagators for Sea Surface Height Forecasts Using Koopman Autoencoders

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M²LInES annual meeting 2024

<https://m2lines.github.io/>

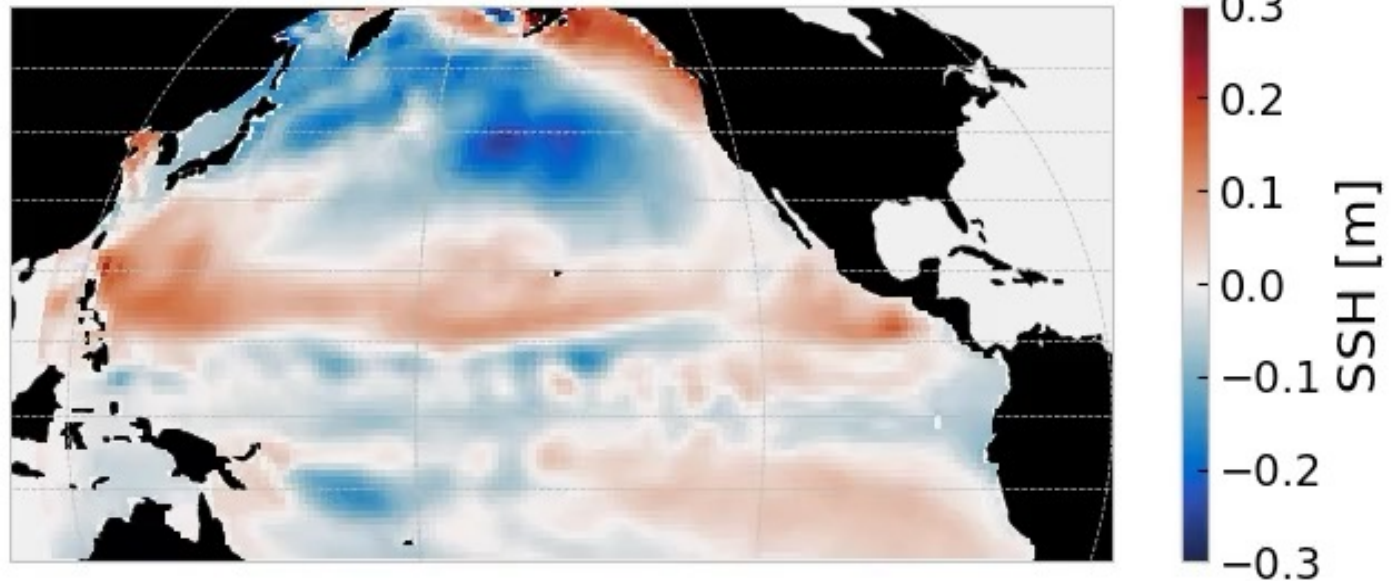




Numerous sources of uncertainty for sea surface height forecasts

Simulated SSH anomalies in CESM2-LE

time = 2021-01-01 00:00:00



Sea level components

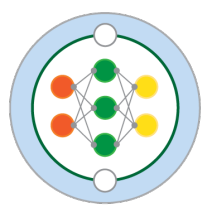
$$\eta = \frac{p_b - p_a}{\rho_0 g} - \frac{1}{\rho_0} \int_{-H}^0 \rho \, dz$$

manometric

surface wind stress
inverse barometer
Kelvin waves
Rossby waves
advection

steric

surface heat flux
heat transport



Statistical-dynamical approaches: Linear inverse modeling (LIM)

- Assume system state is governed by a linear, stochastic dynamical system:

$$\frac{dz}{dt} = \mathbf{A}z + \xi; \quad \xi_i \sim N(0, \sigma_i^2)$$

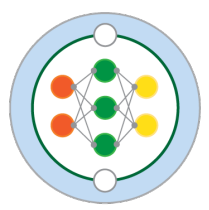
- Best estimate for $\mathbf{A}$:

$$\mathbf{A} = \frac{1}{\tau_0} \log[\mathbf{C}(\tau_0)\mathbf{C}(0)^{-1}]$$

where the covariance matrix $\mathbf{C}$ is given by $\mathbf{C}_{ij}(t + \tau) = \langle z_i(t + \tau)z_j(t) \rangle$.

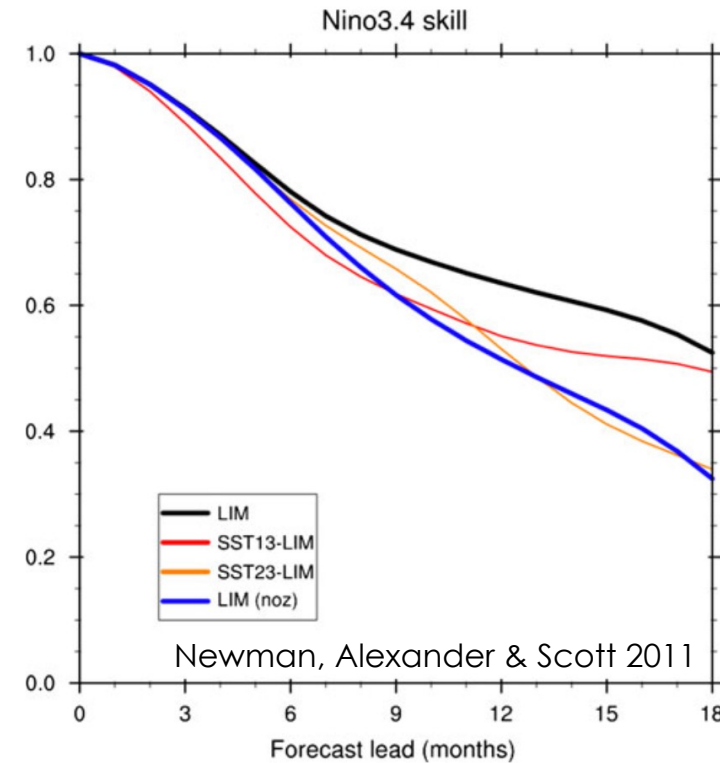
- Forecasts at lag τ are given by

$$\hat{z}_{n+\tau} = e^{\mathbf{A}\tau} z_n$$



Challenges with LIM

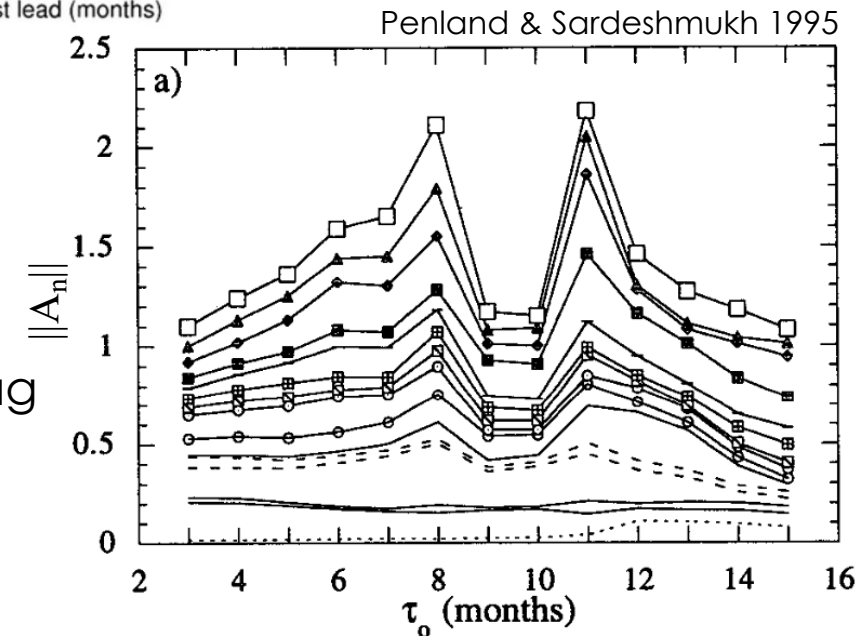
- **Dimensionality reduction:**
Computing $\mathbf{C}(\tau_0)\mathbf{C}(0)^{-1}$ requires low-dimensional state variable z
- **Nonlinear dynamics:** highly nonlinear dynamics may not be well-represented using a linear propagator

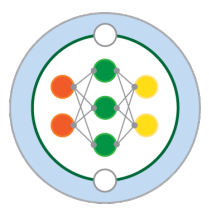


Worse long-term predictions with more PCs than with fewer PCs!!

Newman, Alexander & Scott 2011

Linear propagator depends on time lag

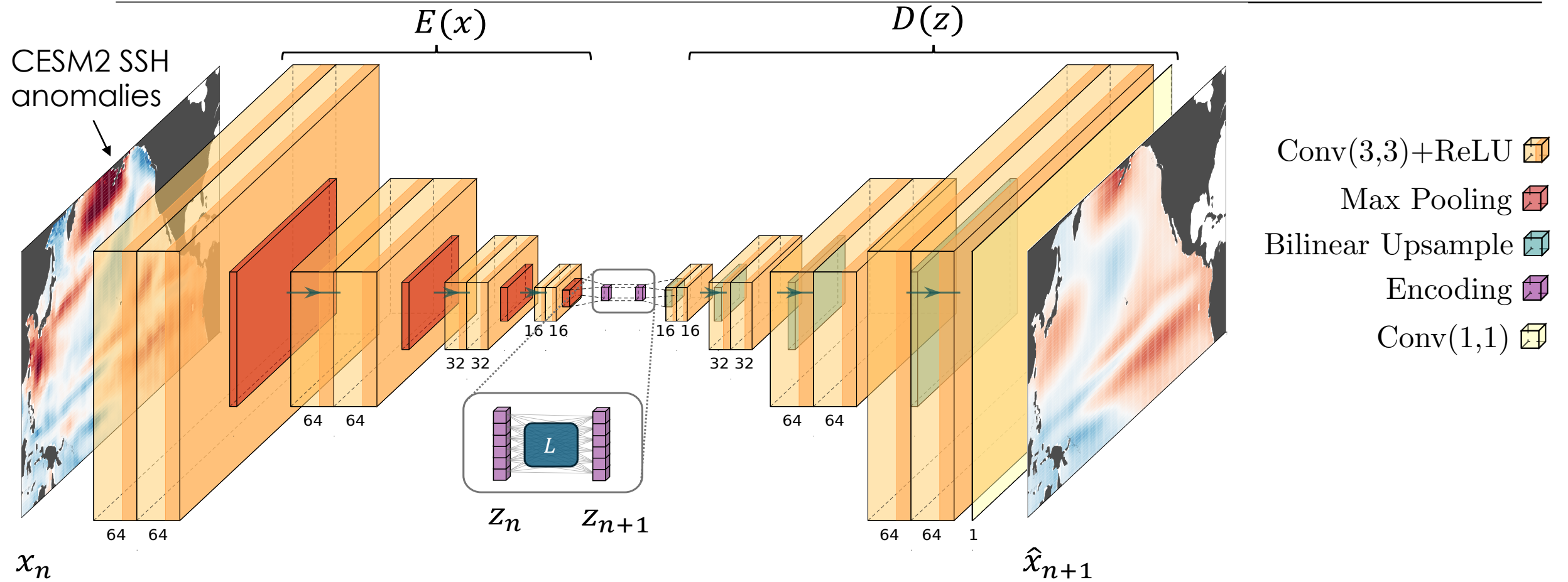




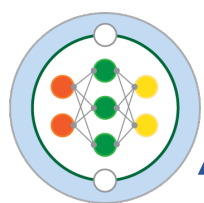
Goals

- Can we perform dimensionality reduction in a way that results in better forecasts of SSH on daily-to-interannual timescales?
- Can we ensure that nonlinear dynamics are well-modelled by a linear propagator?

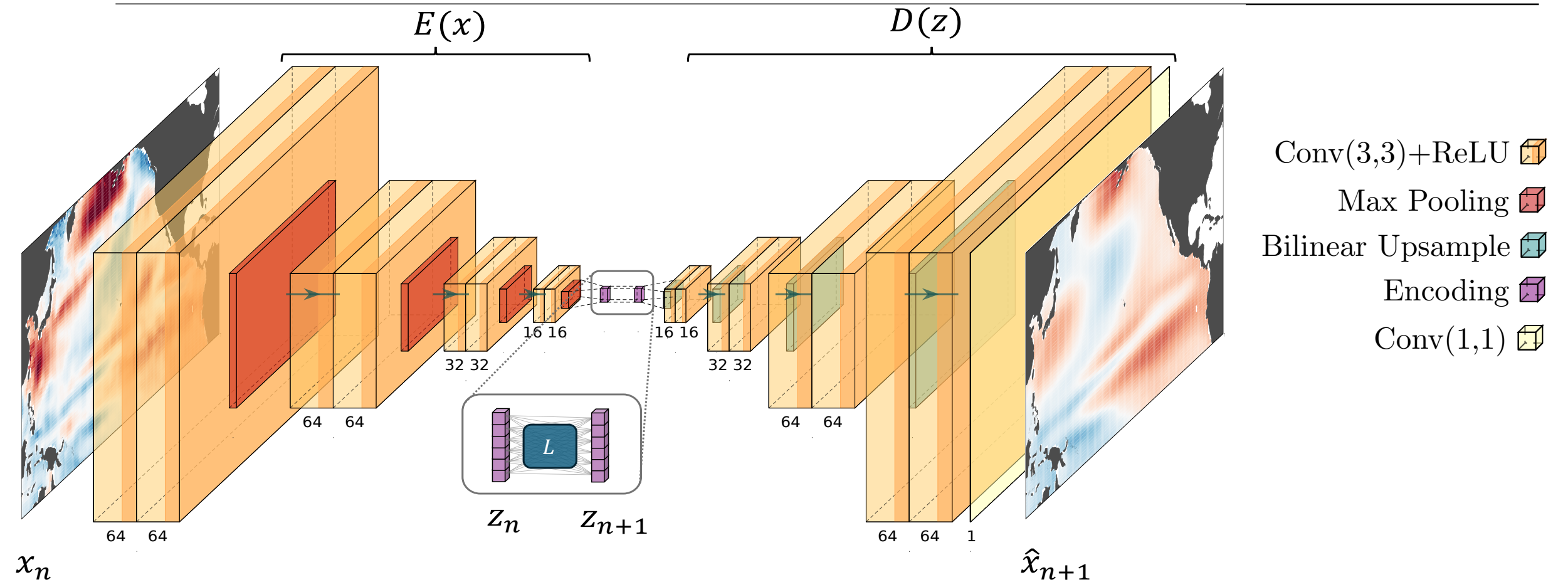
Approach: Koopman Autoencoder



Timestepping and dimensionality reduction are learned together!



Approach: Koopman Autoencoder



Loss functions

1. Reconstructions

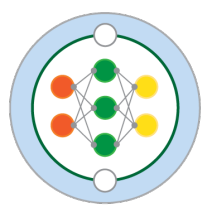
$$\mathcal{L}_{\text{reconst}}(x_n) = \|x_n - D(E(x_n))\|_{2,w}^2$$

2. Predictions

$$\mathcal{L}_{\text{pred}}(x_n, \dots, x_{n+k}) = \frac{1}{k} \sum_{\ell=1}^k \|x_{n+\ell} - D(L^\ell E(x_n))\|_{2,w}^2$$

3. Linearity

$$\mathcal{L}_{\text{linear}}(x_n, x_{n+1}) = \|LE(x_n) - E(x_{n+1})\|_2^2$$



Baselines

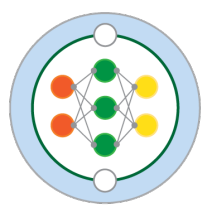
Compare Koopman Autoencoder to baselines when dimensionality reduction and propagators are learned separately:

Dimensionality reduction techniques

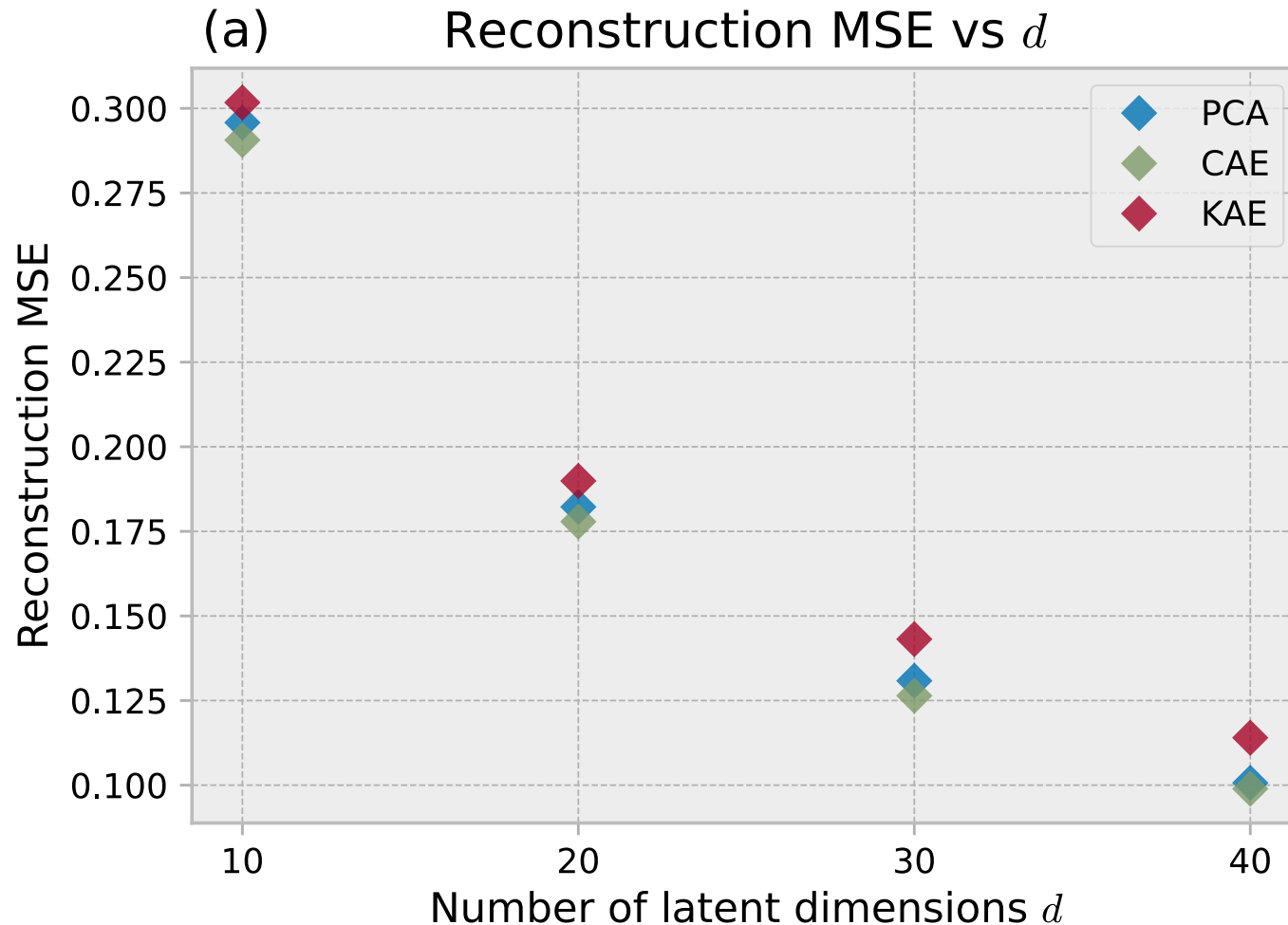
1. Principal Component Analysis (PCA)
2. Convolutional Autoencoder (CAE)
 - Same architecture as the Koopman Autoencoder, but without linear embedding

Latent-space propagators

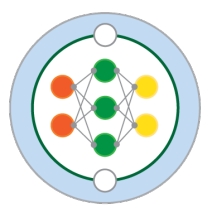
1. Damped persistence (DP)
2. Linear inverse modeling (LIM)



Sensitivity to dimensionality of propagator



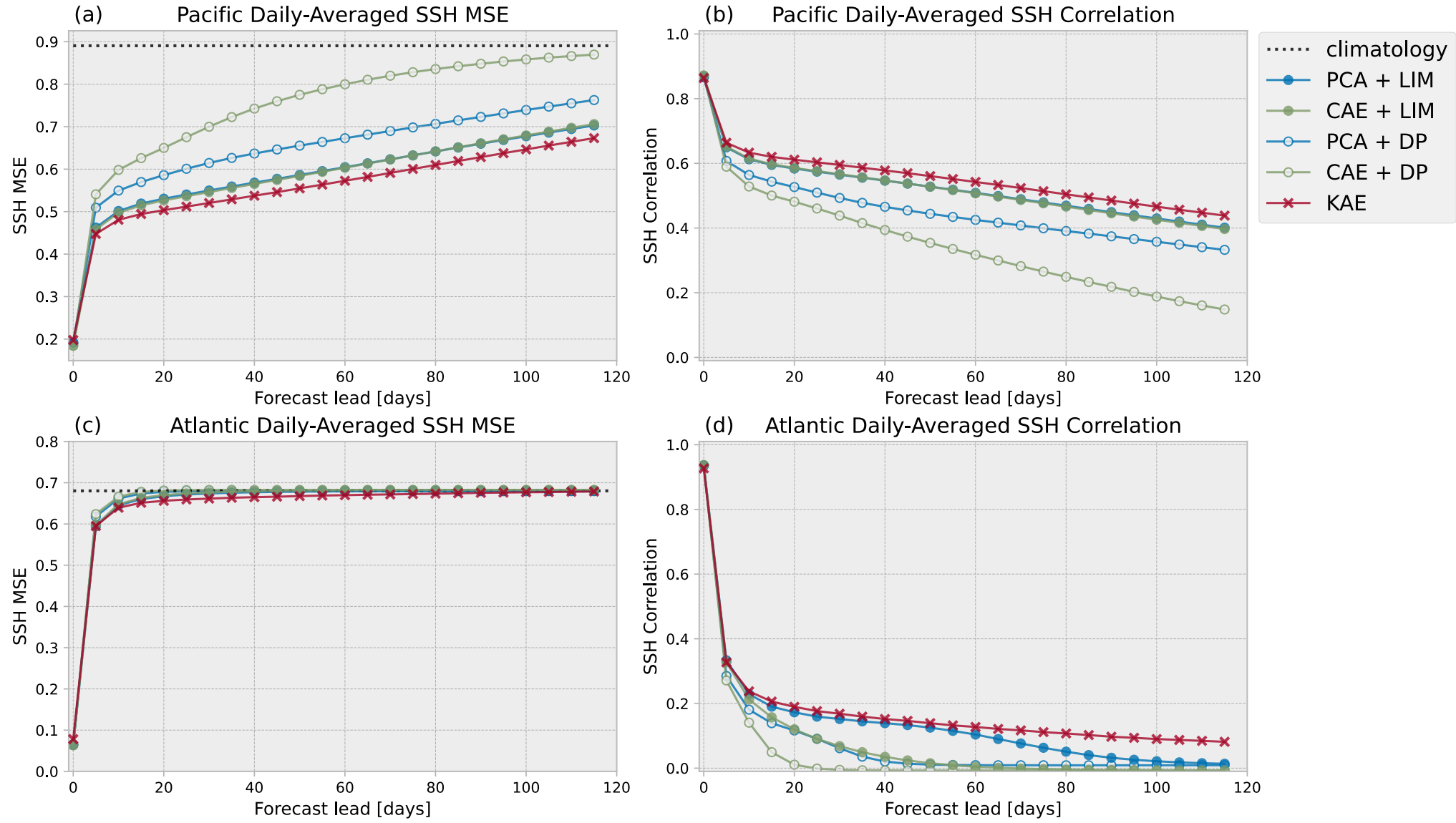
- Reconstruction performance improves with dimensionality
- CAE has lowest reconstruction MSE
- Koopman Autoencoder has the lowest reconstruction error

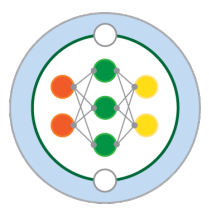


Performance metrics

Colors: dimensionality reduction technique
Markers: propagator type

Prediction metrics, $d=20$

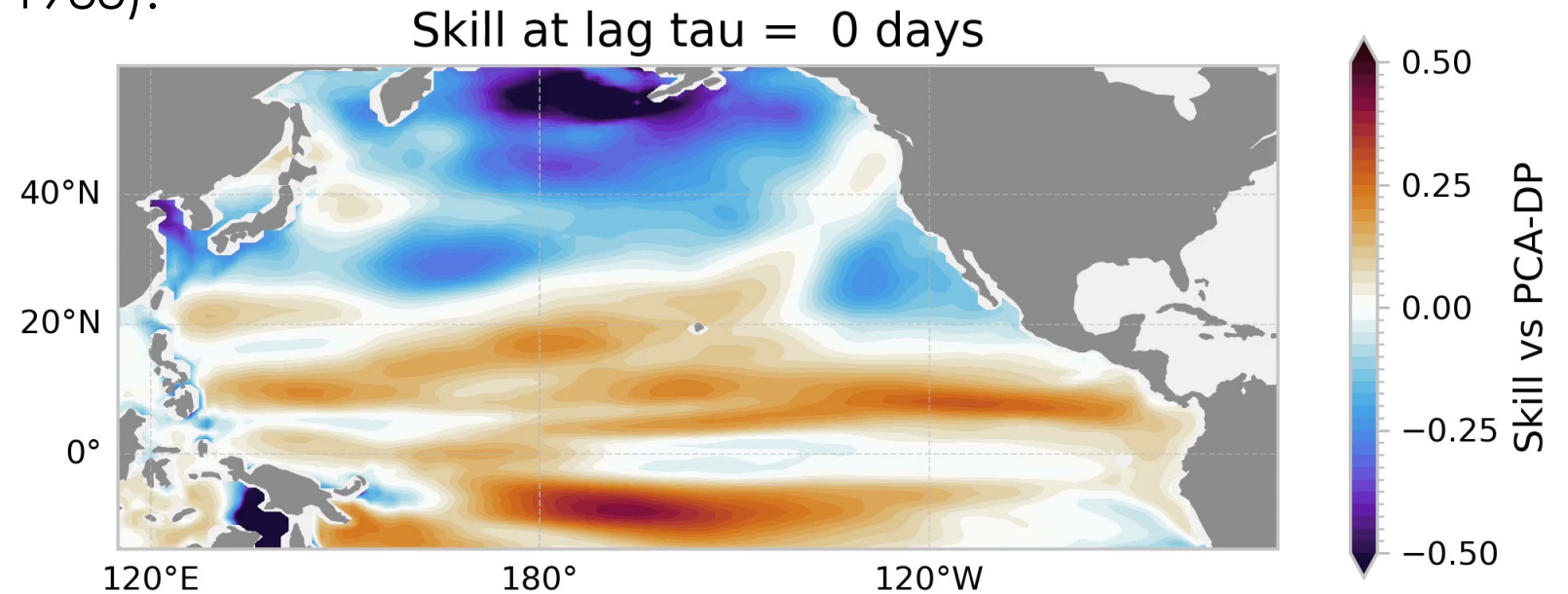




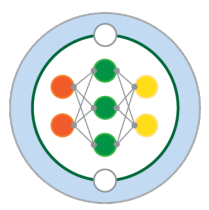
Regions of skill

Skill of Koopman Autoencoder
relative to PCA+DP (Murphy, 1988):

$$SS = 1 - \frac{MSE_{KAE}}{MSE_{PCA+DP}}$$



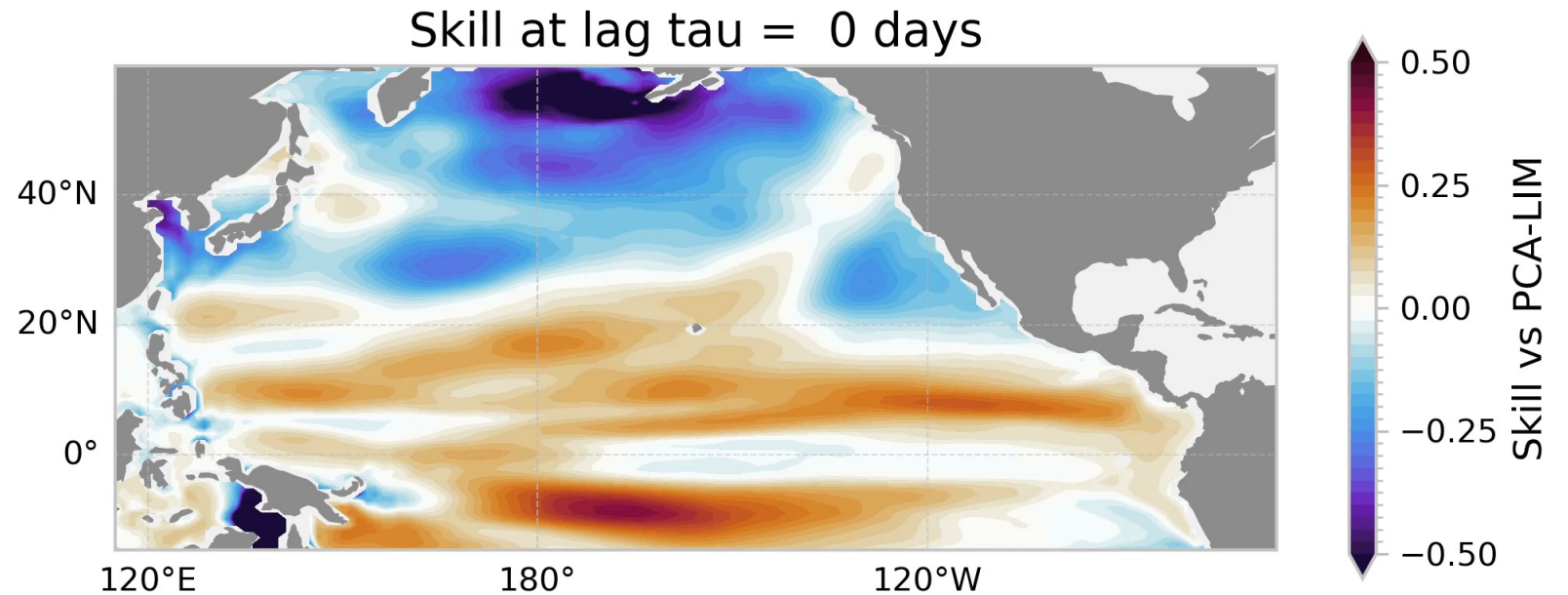
$SS \approx 1$: high skill relative to baseline
 $SS \approx 0$: low skill

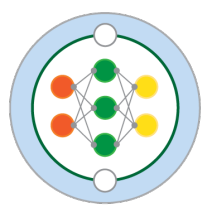


Regions of skill

Skill of Koopman Autoencoder
relative to PCA+LIM:

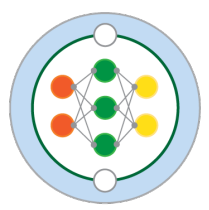
$$SS = 1 - \frac{MSE_{KAE}}{MSE_{LIM}}$$





Conclusions

- Learning dimensionality reduction and a propagator simultaneously in the Koopman Autoencoder results in a better propagator for SSH forecasts
- Reconstruction skill of the model identifies regions in which improved representation can result in better regional predictions



References

1. Gregory, J. M., Griffies, S. M., Hughes, C. W., Lowe, J. A., Church, J. A., Fukimori, I., . . . others (2019). Concepts and terminology for sea level: Mean, variability and change, both local and global. *Surveys in Geophysics*, 40, 1251–1289.
2. Penland, C., & Sardeshmukh, P. D. (1995). The optimal growth of tropical sea surface temperature anomalies. *Journal of Climate*, 8(8), 1999–2024.
3. Lusch, B., Kutz, J. N., & Brunton, S. L. (2018). Deep learning for universal linear embeddings of nonlinear dynamics. *Nature Communications*, 9(1), 4950.
4. Koopman, B. O. (1931). Hamiltonian systems and transformation in Hilbert space. *Proceedings of the National Academy of Sciences*, 17(5), 315–318.